

Inert Load Interlock

Requested, applied, and fault are different facts

Five LEDs make a bounded software policy visible without energizing a load.

Lesson	023 — component	Time	150–210 minutes
Prerequisites	Lessons 001–022	Board	Arduino Mega 2560
Primary evidence	D13, D38, D40, D41, D12	Status	physical acceptance open
Energy class	E1: USB and resistor-limited LEDs only		

Hard boundary. Fan, pump, and heater are names for three LED channels. Do not connect a relay, motor, pump, heater, transistor board, external load supply, or mains circuit. Command admission is a software fact, not power evidence. ADK is not a functional safety controller.

1 Mission

Build an observable chain with no energetic load:

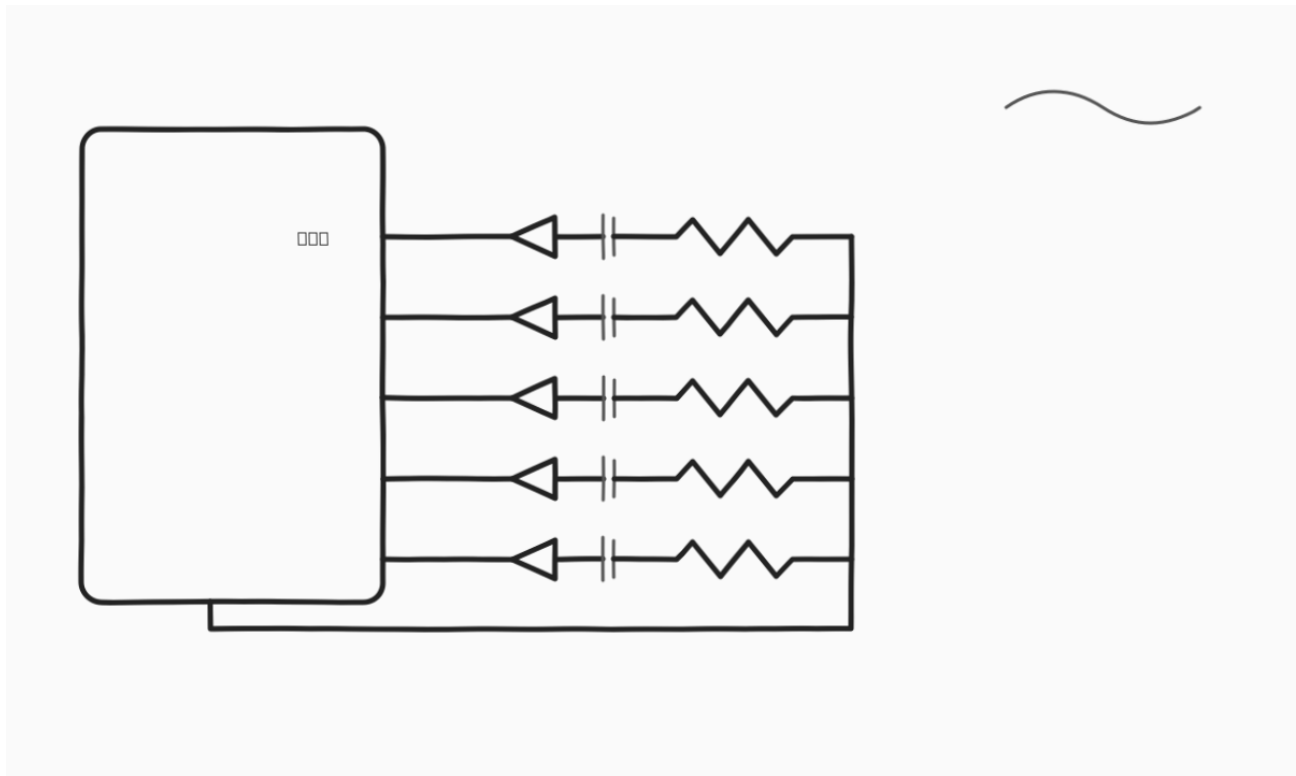
request → admission policy → one applied intent → LED evidence.

At the end you should be able to explain why these statements differ:

1. a channel was requested;
2. policy admitted one active intent;
3. an indicator was commanded;
4. the indicator visibly responded; and
5. a real load received power.

This lesson can establish the first four only. The fifth is deliberately outside the circuit.

2 Orientation and wiring

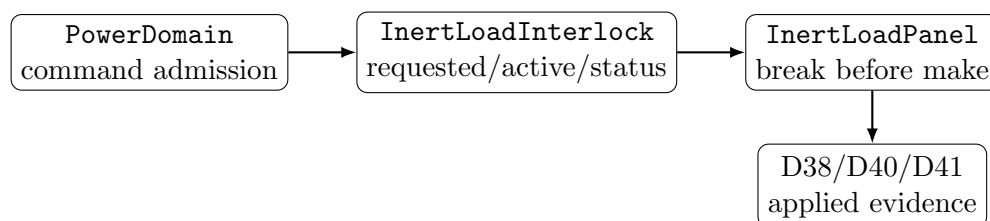


Alternative text: a pencil-style Mega 2560 orientation drawing. D13 shows a request pulse; D40, D38, and D41 show applied fan, pump, and heater intent; D12 shows a latched fault. Each external LED returns through its own resistor to the common ground. No load or relay is connected.

Meaning	Pin	Connection
Requested intent	D13	Built-in LED; no external wire
Applied pump	D38	D38 to 220 Ω , LED anode; cathode to GND
Applied fan	D40	D40 to 220 Ω , LED anode; cathode to GND
Applied heater	D41	D41 to 220 Ω , LED anode; cathode to GND
Latched fault	D12	D12 to 1 k Ω , LED anode; cathode to GND

Unpowered inspection. Remove USB. Confirm LED polarity, one resistor per LED, a common Mega GND, and no connection to a relay or powered load. Record resistor measurements: D38 _____, D40 _____, D41 _____, D12 _____.

3 The object story



`InertLoadInterlock` owns no pin. It records one requested selection and at most one active intent. Each

IndicatorPump owns one indicator endpoint. **InertLoadPanel** borrows the three outputs in fan, pump, heater order and renders the applied selection; it cannot outlive them. These relationships are not evidence that a real load is safe.

3.1 Lifecycle

Construction is inert. Initialization establishes **None**. Repeated initialization is idempotent. Shutdown selects all-off and releases owned endpoints. Destruction calls shutdown and remains safe during caller exception unwinding, although ADK itself does not throw.

A revoked admission causes **HardwareFailure**, clears active intent, and leaves the failure sticky. Restoring admission does not silently retry. Recovery is explicit: shutdown, correct the external condition, initialize, then issue a new request.

3.2 One bounded status value

```
const adk::Status status = interlock.select (requestedLoad);

if (!status.ok ())
{
    observeFailure (status);
}
```

Pass the complete value to the observation boundary. Use **error()** only when policy needs the underlying code. **transient()** classifies a cause; it never authorizes an unbounded retry.

4 Read the snapshot

Requested	Active	Status	Meaning
Pump	Pump	Ok	Pump indicator intent was admitted
Heater	Heater	Ok	Prior channel cleared before heater applied
None	None	Ok	Deliberately inactive
Pump	None	HardwareFailure	Request recorded; admission failed closed
Invalid value	unchanged	InvalidArgument	Prior valid state preserved

The request pulse on D13 establishes that a selection attempt occurred. Exactly one of D38, D40, and D41 renders **active**. D12 renders a reported fault. Do not collapse those observations into a single “working” light.

5 Experiment A: mutual exclusion

Predict. Write the expected applied LED after each request.

Step	Request	D13	Applied LED	Prediction
1	Fan	pulse	D40	_____
2	Pump	pulse	D38	_____
3	Heater	pulse	D41	_____
4	None	pulse	none	_____

Observe. Watch the request pulse before the applied channel changes. At no instant should two applied indicators remain illuminated.

Record one complete sequence: _____.

Interpret. Does the sequence support break-before-make? Why?

_____.

6 Experiment B: admission fault

The deterministic host test revokes the borrowed domain while an intent is active. The Mega teaching sketch may expose the same scenario only through a fixed, documented simulation; no power path is interrupted.

1. Predict the next requested, active, and status fields.
2. Observe D13, all three applied LEDs, and D12 separately.
3. Confirm requested may name a load while active remains **None**.
4. Restore admission without restarting. Confirm the fault stays sticky.
5. Restart the lifecycle before making a fresh request.

Moment	Requested evidence	Applied evidence	Fault evidence
Before revoke	_____	_____	_____
After revoke	_____	_____	_____
Admission restored	_____	_____	_____
After lifecycle restart	_____	_____	_____

Interpretation limit. All-off LEDs support fail-closed software intent. They do not establish galvanic isolation, contact position, load voltage, or removal of stored energy.

7 Fault laboratory

Injected condition	Expected public result	Expected evidence
Selection before initialize	NotInitialized	no applied LED
Unknown enum value	InvalidArgument	prior valid selection unchanged
Admission revoked	transient HardwareFailure	active all-off; D12 latched
Old channel cannot turn off	HardwareFailure	best-effort all-off
New channel cannot turn on	HardwareFailure	best-effort all-off
Repeated shutdown	Ok cleanup behavior	every applied LED off
Destruction while active	no throw, no retained claim	pins released

For each row, distinguish the returned **Status**, the snapshot, the recorded endpoint trace, and the physical observation. One cannot substitute for another.

8 Shutdown proof

Resource acquisition and safe electrical state are separate checks.

1. **Acquire:** predict a busy indicator pin causes transactional rollback. Host evidence should show every earlier claim released.
2. **Inactive:** predict each applied LED is dark before release.

3. **High impedance:** with the circuit and meter arranged unpowered, observe D38/D40/D41 after shutdown relative to named GND.
4. **Reset:** do not infer safe state from a destructor. Remove USB power before touching the circuit.

9 Deterministic proof

The host suite should exhaust transitions among **None**, **Fan**, **Pump**, and **Heater**; initialization failures at every endpoint; old-off and new-on failures; invalid input; asynchronous admission revocation; sticky faults; explicit recovery; repeated shutdown; and destruction.

The key trace property is:

$$\forall t, \quad [\text{fan}(t)] + [\text{pump}(t)] + [\text{heater}(t)] \leq 1.$$

Replay the same admission and request trace twice. Requested, active, status, and endpoint records must match exactly.

10 Exercises

1. Draw a state table that distinguishes **requested** from **active**. Why is one field insufficient?
2. Explain why **transient()** must not automatically restart a failed selection.
3. Add a host-only fault at each panel write. Prove the all-off pass is attempted without claiming that it succeeds.
4. Design a two-LED code for requested versus applied without using color. State which information is lost.
5. Explain why a real relay adapter belongs to a different energy class and needs independent power-removal evidence.
6. Explain how **PanelPumpOutput** borrows **InertLoadPanel**, adapts only its pump channel, and why it must not outlive the panel in Lesson 024.

11 CLI workflow

```
make host-test
make arduino-Lesson023InertLoadInterlock
make size-Lesson023InertLoadInterlock
make lessons
make lessons-check
make site-check
```

Compilation, deterministic replay, and PDF generation are software evidence. They leave every physical row below open.

12 Physical acceptance record — intentionally blank

Field	Record
Date, person, commit	
Mega board/revision	
Arduino CLI/core versions	
USB supply and measured rail	
Instrument and identifier	
Measured resistor values	
Unpowered wiring inspection	
Acquisition prediction/observation	
Normal sequence prediction/observation	
Mutual-exclusion observation	
Fault prediction/observation	
Shutdown LED observation	
High-impedance observation	
Reset and power-removal observation	
Deviation and stop condition	
Result: ☐ pass ☐ fail ☐ not run. Until a named person fills this record from a real bench, the lesson remains host verified at most.	

13 References and next step

- Canonical Lesson 023 example
- Interface and tests: https://github.com/spincyc/adk/blob/main/src/inert_load_interlock.h and https://github.com/spincyc/adk/blob/main/tests/test_inert_load_interlock.cpp
- ATmega2560: <https://www.microchip.com/en-us/product/atmega2560>
- ADK safety model: https://github.com/spincyc/adk/blob/main/docs/SAFETY_MODEL.md

Lesson 024 composes this bounded inert output with calibrated moisture samples and deterministic watering time. It still uses an LED as the reference pump.